

ES -116

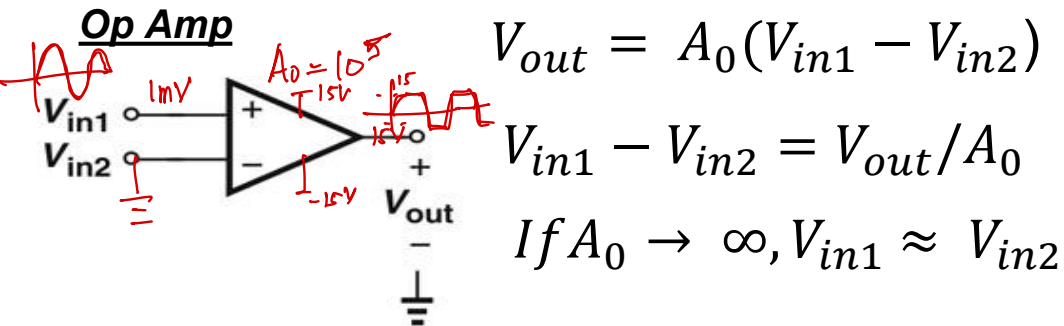
Principles and Applications of Electrical Engineering

Semester II, 2025-26

Utilities of OpAMP: Signal conditioning

Recap

Op Amp

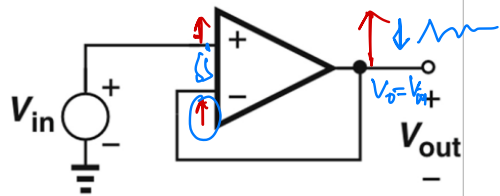


Virtual short

Due to infinite gain of op amp, V_{in2} appears to same as V_{in1} .

Power of (negative) feedback to control the gain

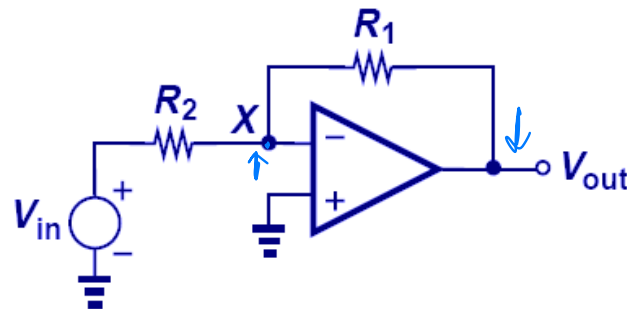
$A_0 = 10^5$ (ideally infinity)



Infinite A_0 : $\frac{V_{out}}{V_{in}} = 1$

Popularly known as voltage buffer/follower/unity-gain circuit

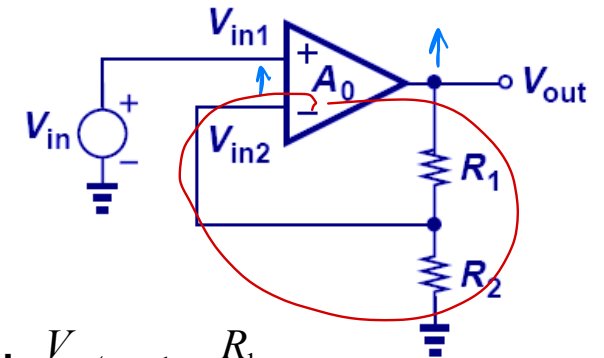
Inverting Op Amp



Infinite A_0 : $\frac{V_{out}}{V_{in}} = -\frac{R_1}{R_2}$

Finite A_0 : $\frac{V_{out}}{V_{in}} \approx -\frac{R_1}{R_2} \left[1 - \frac{1}{A_0} \left(1 + \frac{R_1}{R_2} \right) \right]$

Non-Inverting Op Amp



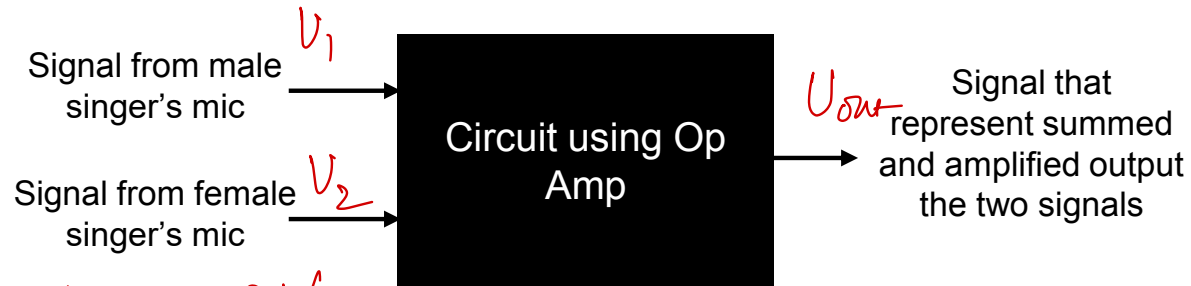
Infinite A_0 : $\frac{V_{out}}{V_{in}} = 1 + \frac{R_1}{R_2}$

Finite A_0 : $\frac{V_{out}}{V_{in}} \approx \left(1 + \frac{R_1}{R_2} \right) \left[1 - \left(1 + \frac{R_1}{R_2} \right) \frac{1}{A_0} \right]$

An interesting application of Op Amp based circuits

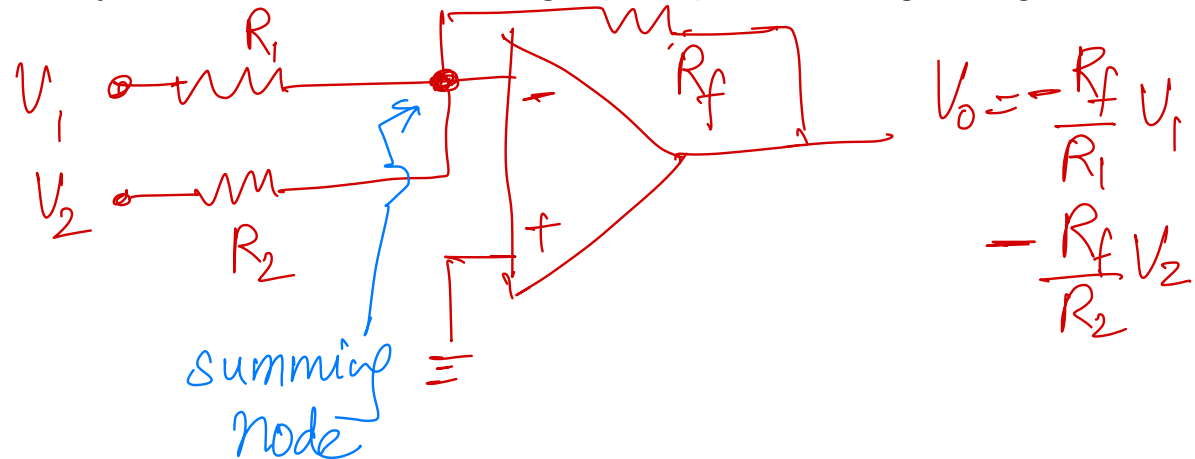


Audio mixer

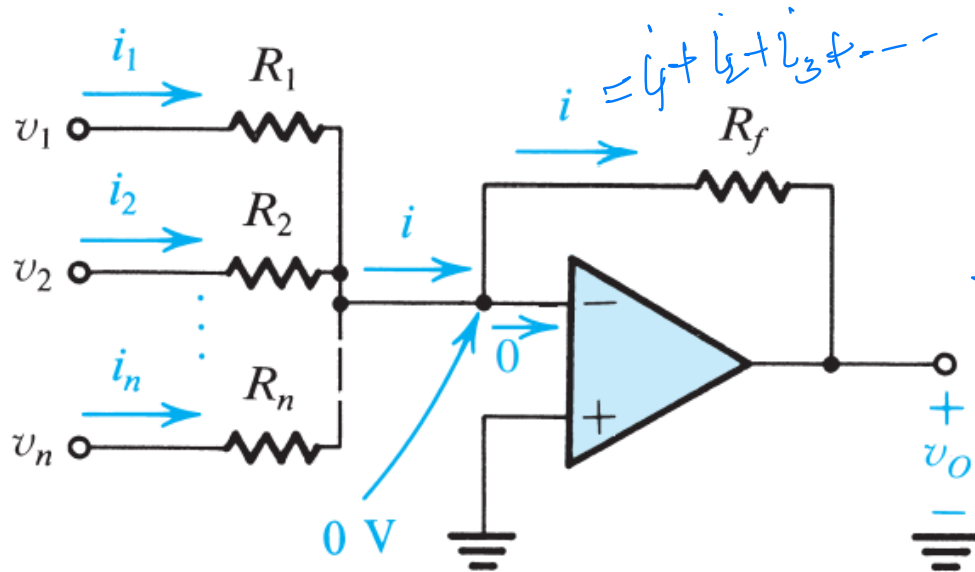


$$V_{out} = A V_1 + B V_2$$

Can you make this circuit using Op Amp in inverting configuration?



Voltage summing amplifier



$$V_o = - \left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} + \dots \right)$$

$$\frac{R_f}{R_1} \approx 10$$

① $R_1 = 10 \Omega, R_f = 100 \Omega$

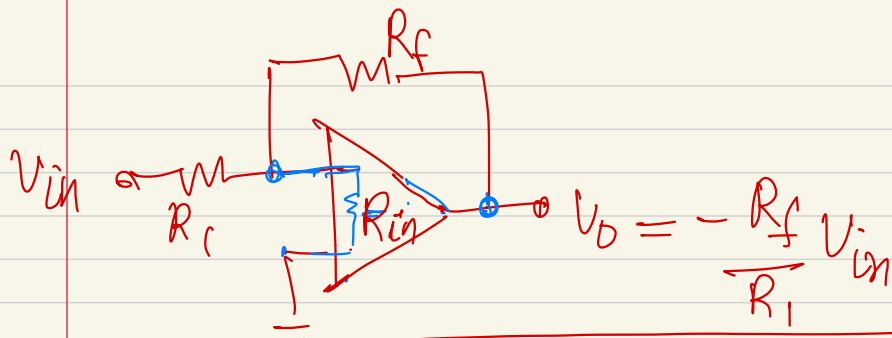
② $R_1 = 100 \Omega, R_f = 1 \text{ k}\Omega$

③ $R_1 = 1 \text{ k}\Omega, R_f = 10 \text{ k}\Omega$

④ $R_1 = 10 \text{ k}\Omega, R_f = 100 \text{ k}\Omega$

Options to get the same gain, how do I decide on R_1 value?

What dictates the choice of $R_1, R_2, \dots, R_n$?

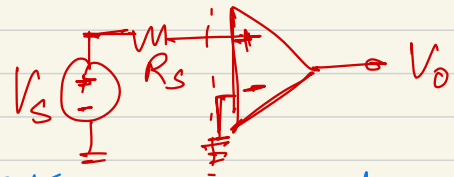
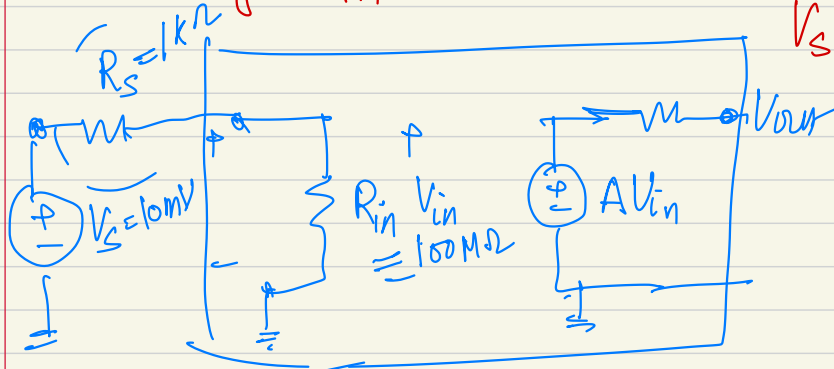


How R_i & R_f values are chosen?

If R_{in} is the internal resistance of op-amp looking inside the inverting terminal of the op-amp, it is typically very high ($10\text{M}\Omega - 10^{12}\Omega$).

That is why we can assume $R_{in} \rightarrow \infty$.

→ Assuming $R_{in} = 100\text{M}\Omega$



input loading

$$V_{in} = V_s \frac{R_{in}}{R_s + R_{in}}$$

$$\therefore R_{in} = 100\text{M}\Omega \text{ (internal resistance of op-amp)}$$

$$\Rightarrow V_{in} = 10\text{mV} \times \frac{100\text{M}}{1\text{k} + 100\text{M}} \approx 10\text{mV}$$

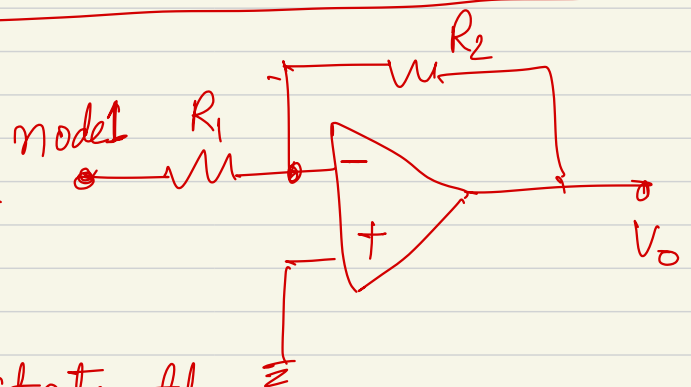
If faulty IC $\rightarrow R_{in} = 1\text{k}\Omega$, $V_{in} \sim 5\text{mV}$

Assuming $R_{in} \rightarrow \infty$

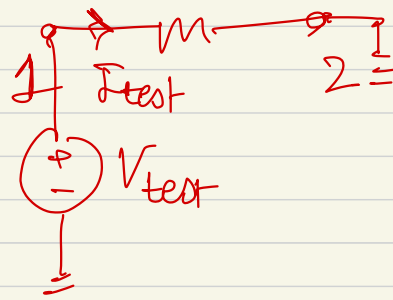
What is input resistance looking node 1?

→ It would dictate the

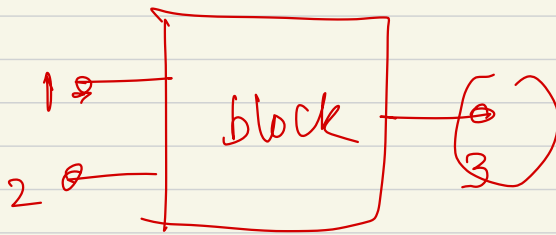
maximum source resistance that you can bear without losing the signal at the source.



Resistor measurement using Multimeter

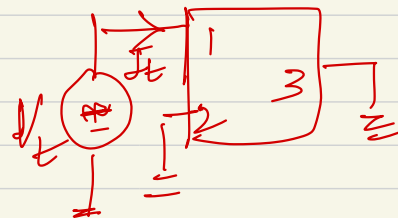


$$\frac{V_{\text{test}}}{I_{\text{test}}} = R$$



$R_1 = ?$
test source to '1'
2 & 3 $\rightarrow$ Ground

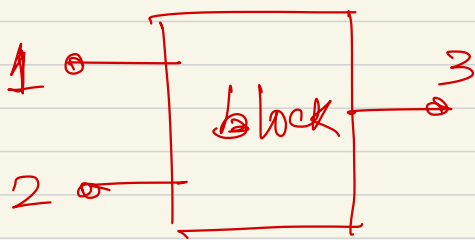
What is R_1 , R_2 , R_3 ?



$$R_1 = \frac{V_t}{I_t}$$

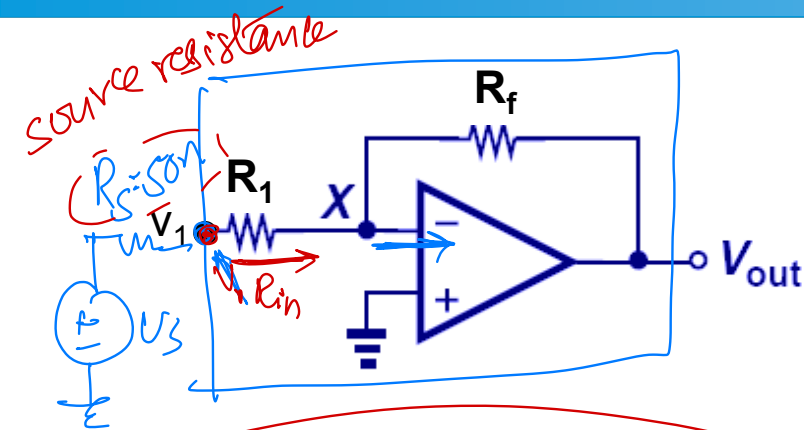


Method to calculate the Resistance looking into a particular node



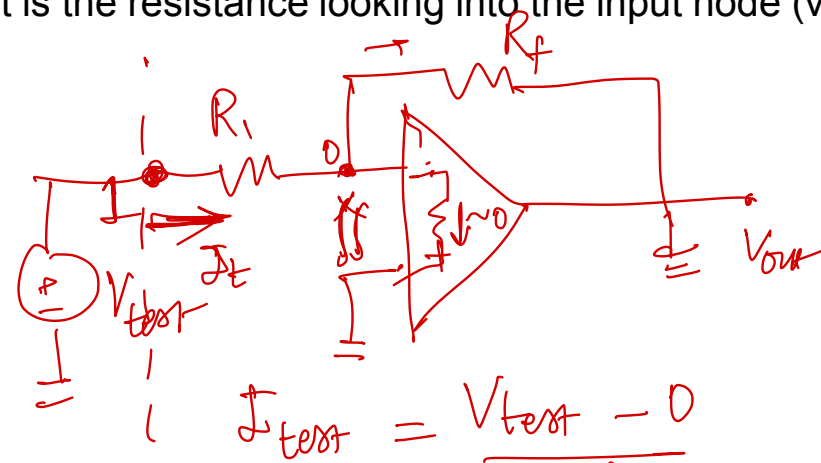
- 1) Apply test source to the node in question. (e.g. '1')
- 2) Deactivate the contribution from other nodes (e.g. Ground at 2 & 3).

How do you choose the value of Input resistance?



Assume a source (e.g. a functional generator) with a source resistance of $50\ \Omega$ connected to the input v_1 , what resistances (R_1 and R_f) can I choose for a gain of -10.

What is the resistance looking into the input node (v_1)?



$$I_{test} = \frac{V_{test} - 0}{R_1}$$

$$R_{in}(\text{node 1}) = \frac{V_{test}}{I_{test}} = R_1$$

→ R_1 should be large enough to transfer the source input without dropping it across R_s .

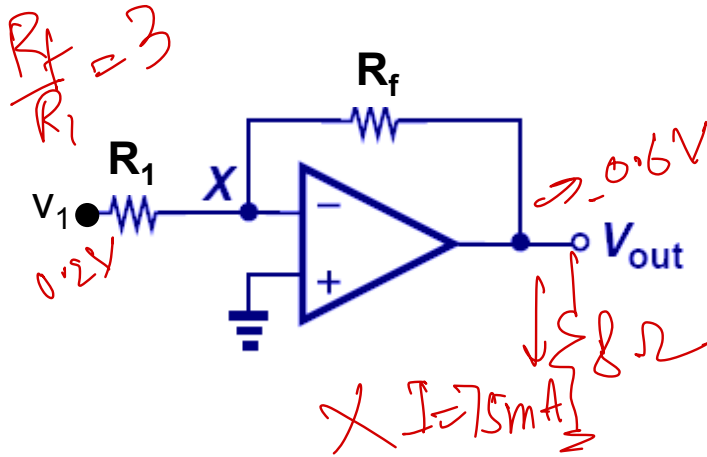
① $R_1 = 10\ \Omega$, $R_f = 100\ \Omega$ X

② $R_1 = 100\ \Omega$, $R_f = 1\ \text{k}\Omega$ X

③ $R_1 = 1\ \text{k}\Omega$, $R_f = 10\ \text{k}\Omega$

④ $R_1 = 10\ \text{k}\Omega$, $R_f = 100\ \text{k}\Omega$

Can you draw any output current that you like?



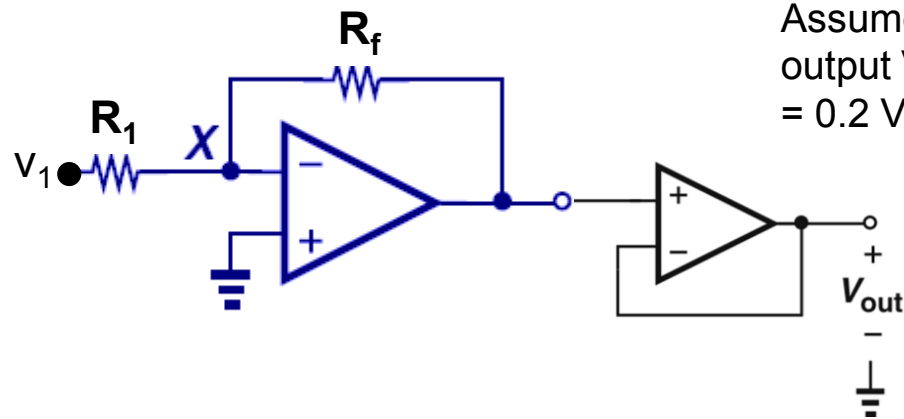
Assume a load (e.g. speaker) with a resistance of $8\ \Omega$ connected to the output V_{out} , would the OpAMP be able to drive it with gain -3. Assume $v_1 = 0.2\text{ V}$.

Typically, there is a strict physical limit on how much current they can push out, usually around 25 mA (for uA741).

The load requires $\frac{0.6V}{8\Omega} = 75\text{ mA}$ current drawn. However, Op-Amp would not be able to supply output currents more than 25 mA. Therefore, V_o can not 0.6 V corresponding to 0.2 V (peak).

The expected 0.6 V signal severely attenuates as the Op-Amp panics → the internal protection circuits kick in → the feedback loop breaks → and the audio sounds like distorted garbage.

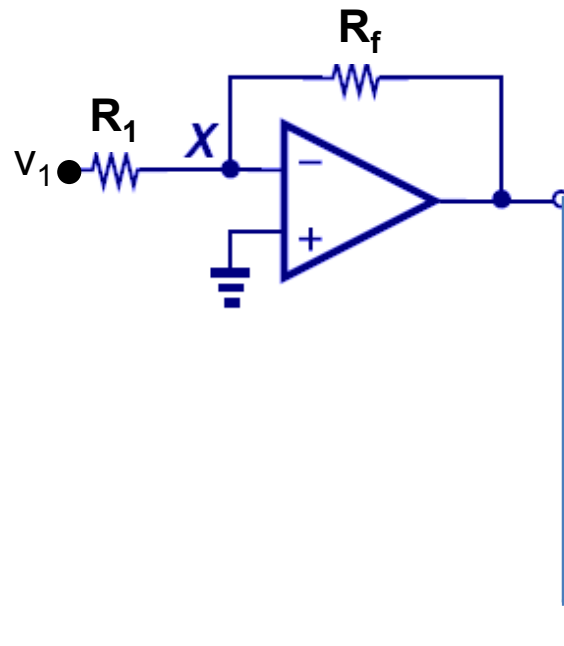
Would voltage follower help?



Assume a load (e.g. speaker) with a resistance of $8\ \Omega$ connected to the output V_{out} , would the OpAMP be able to drive it with gain -3. Assume $v_1 = 0.2\text{ V}$.

- It does separate the output of inverting amplifier, there remove the loading at the output of inverting amplifier.
- But, can it provide the required current or power to the load?

Solution if you driving a low load like Speaker



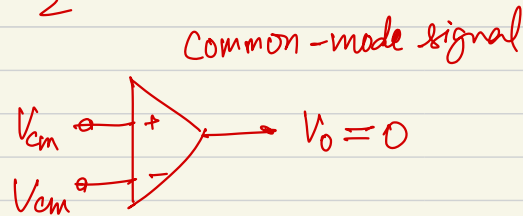
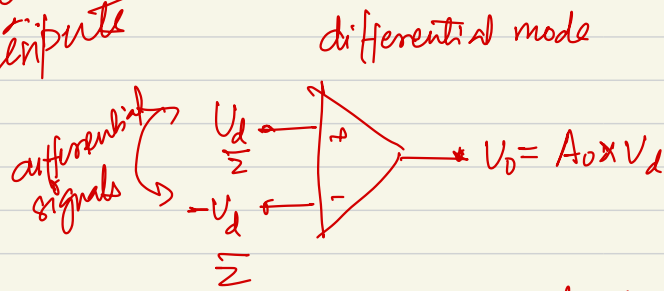
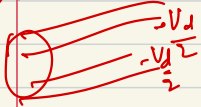
Assume a load (e.g. speaker) with a resistance of $8\ \Omega$ connected to the output V_{out} , would the OpAMP be able to drive it with gain -3. Assume $v_1 = 0.2\text{ V}$.



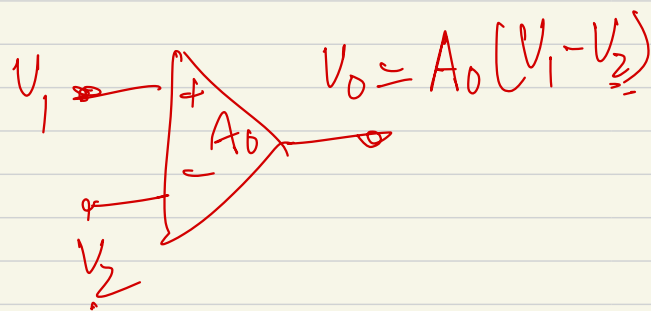
V_{out}

Maximum output power $\sim 0.7\text{ W} - 1\text{ W}$.
i.e. for $8\ \Omega$ load, currents that can be drawn $\sim 300\text{ mA}$

coaxial cable with differential inputs



If U_1 & U_2 are ECG signals acquired by electrodes.

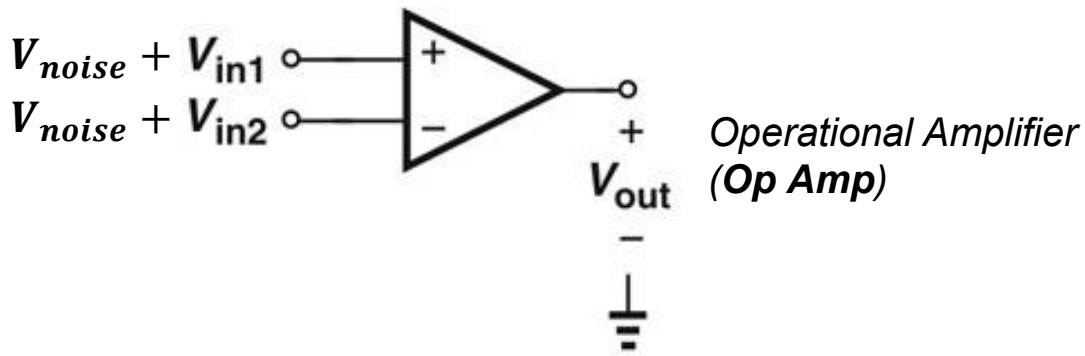


$$U_1 = \underbrace{U_{id}}_2 + U_{noise}, \quad U_2 = \underbrace{-U_{id}}_2 + U_{noise}$$

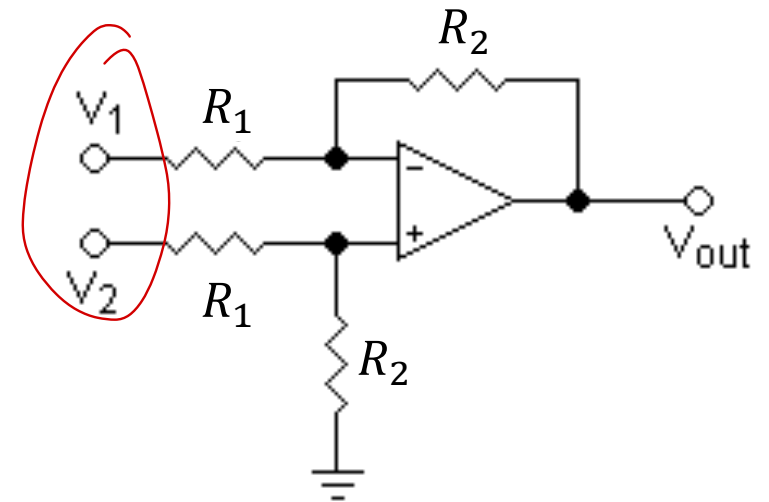
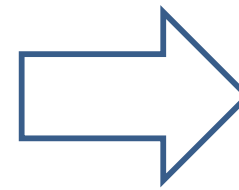
$U_{id} \rightarrow$ differential inputs carrying info. from electrodes
 $U_{noise} \rightarrow$ common-mode signal added to both inputs. It is added to both inputs as it is a random signal which is converted to additive voltage signal from RMS noise power.

When I require to use both inputs of OpAMP

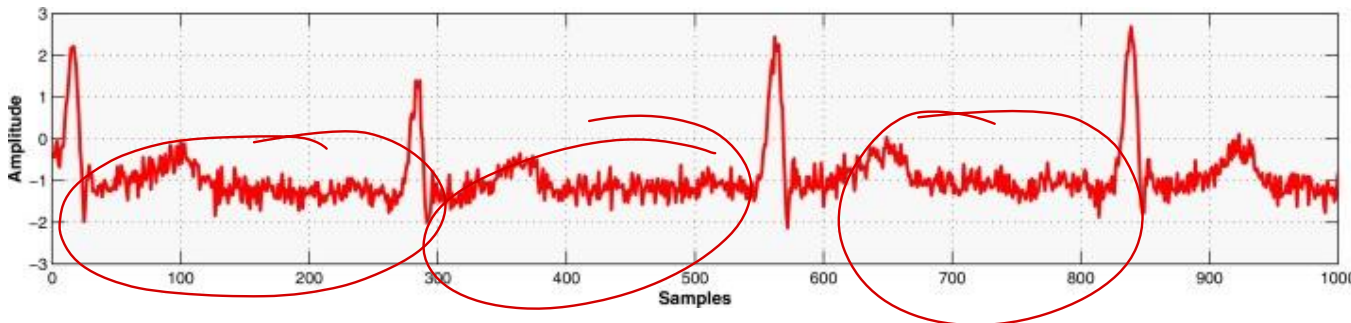
$$V_{out} = A_0(V_{in1} - V_{in2})$$



With feedback



- To reject the common signal such as noise present in the input signal (example shown: ECG signal).

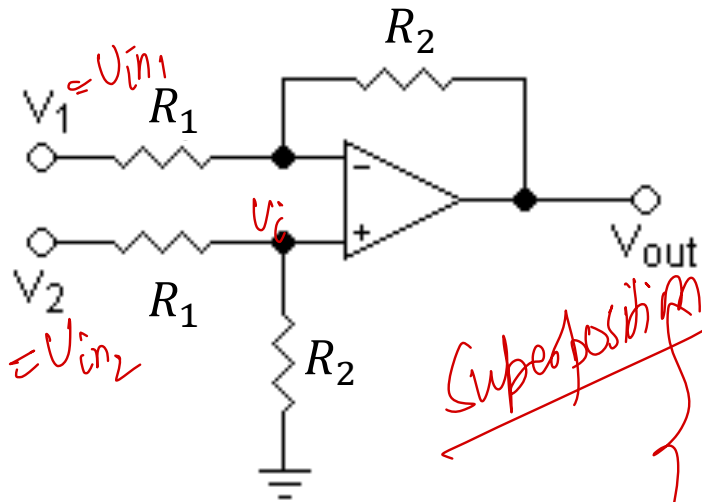


Analyse the given circuit when noise gets added to the inputs:

$$V_1 = V_{in1} + V_{noise}$$

$$V_2 = V_{in2} + V_{noise}$$

Analysis of the difference amplifier



Analyse the given circuit when noise gets added to the inputs:

$$V_1 = V_{in1} + V_{noise}$$

$$V_2 = V_{in2} + V_{noise}$$

$$\textcircled{1} \Rightarrow V_{out} = V_{out1} + V_{out2}$$

$$+ \textcircled{2} \quad V_{out} = \frac{R_2}{R_1} (V_{in2} - V_{in1})$$

Approach: Let's use superposition theorem

Superposition

from $V_{in1} \Rightarrow V_{in2} \Rightarrow$

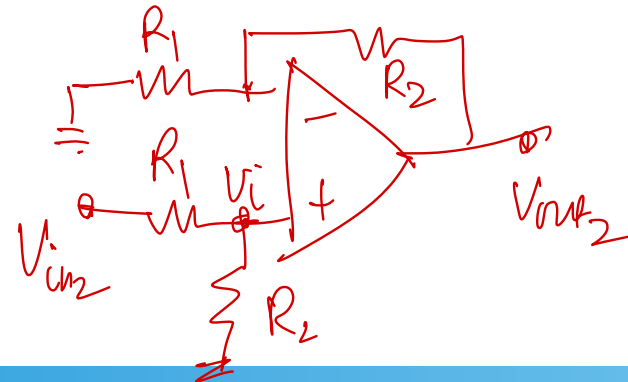
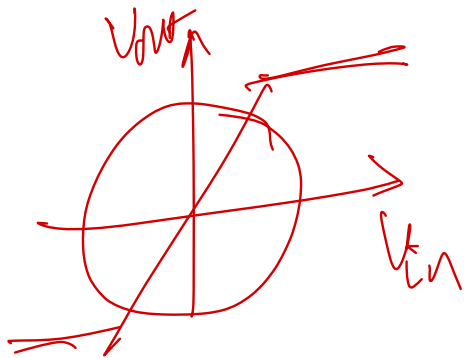
from V_{in2} , $V_{in1} = 0$

$$V_{out1} = -\frac{R_2}{R_1} V_{in1} \quad \text{---} \textcircled{1}$$

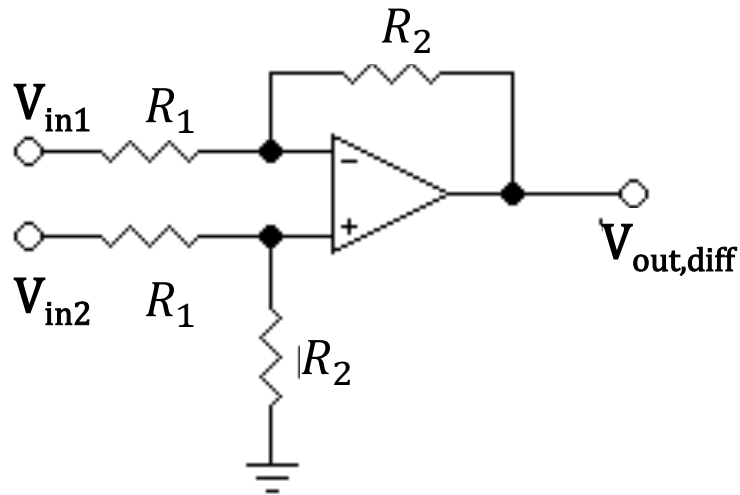
$$V_{out2} = \left(1 + \frac{R_2}{R_1}\right) V_i$$

$$= \left(1 + \frac{R_2}{R_1}\right) \cdot \frac{R_2}{R_1 + R_2} V_{in2}$$

$$= \frac{R_2}{R_1} V_{in2} \quad \text{---} \textcircled{2}$$



Solving for differential input signal



Consider V_{in1} and V_{in2} , and find out $V_{out,diff}$?

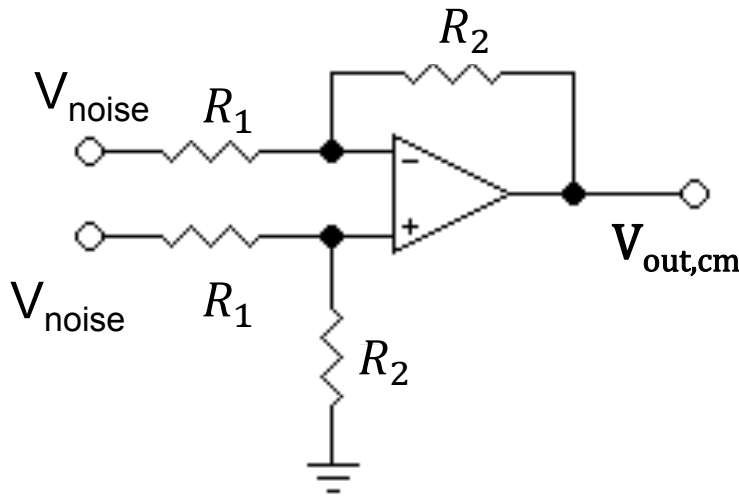
$$V_{in1} = -\frac{V_{id}}{2}$$

$$V_{in2} = +\frac{V_{id}}{2}$$

$$\Rightarrow V_{o,diff} = \frac{R_2}{R_1} (V_{in2} - V_{in1}) = \frac{R_2}{R_1} V_{id}$$

$$\Rightarrow A_d = \frac{V_{o,diff}}{V_{id}} = \frac{R_2}{R_1}$$

Solving for common input signal

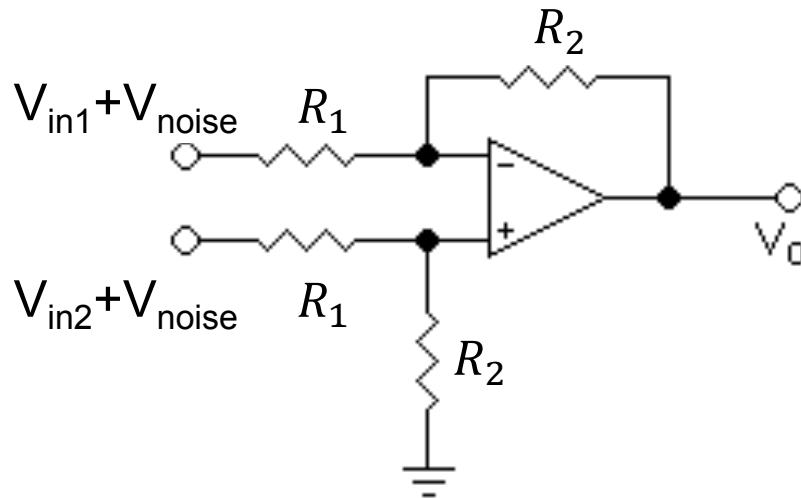


Now, consider V_{noise} as the input and find out $V_{\text{out,cm}}$?

$$\begin{aligned}
 V_{\text{out,cm}} &= \frac{R_2}{R_1} (V_2 - V_1) \\
 &= \frac{R_2}{R_1} (V_{\text{noise}} - V_{\text{noise}}) = 0
 \end{aligned}$$

$$A_{\text{cm}} = \frac{V_{\text{out,cm}}}{V_{\text{noise}}} = 0$$

Total V_{out}



$$V_{out} = V_{out,diff} + V_{out,cm}$$

$$= A_d \cdot U_{id} + A_{cm} \cdot U_{cm}$$

=

CMRR

The efficacy of a difference amplifier is measured by the degree of its rejection of common-mode signals quantified by a measure known as the **common-mode rejection ratio (CMRR)**.

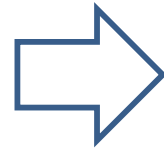
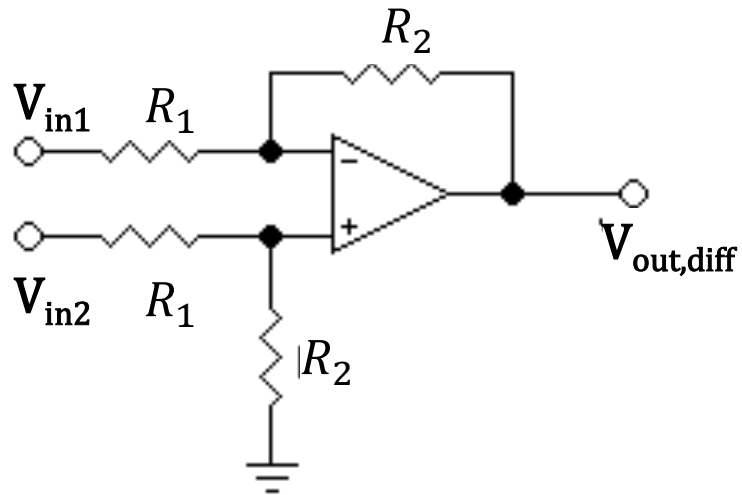
$$A_d = \frac{V_{o,diff}}{V_{id}}$$

$$A_{cm} = \frac{V_{o,cm}}{V_{cm}}$$

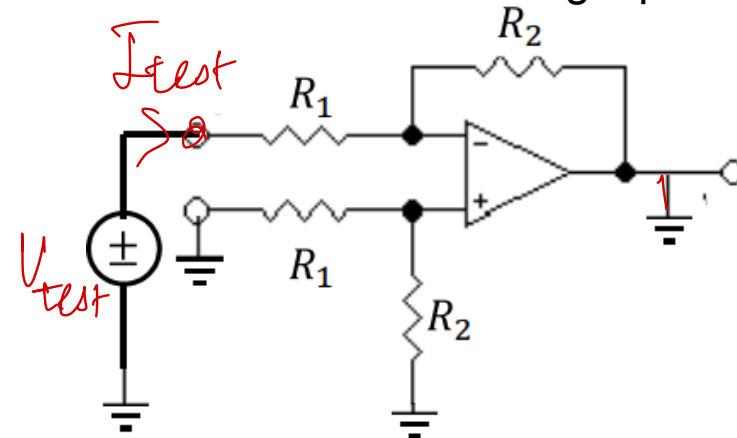
$$CMRR = 20 \log_{10} \left| \frac{A_d}{A_{cm}} \right|$$

ideal: $A_{cm} \rightarrow 0$, $CMRR \rightarrow \infty$

Issue of input impedance



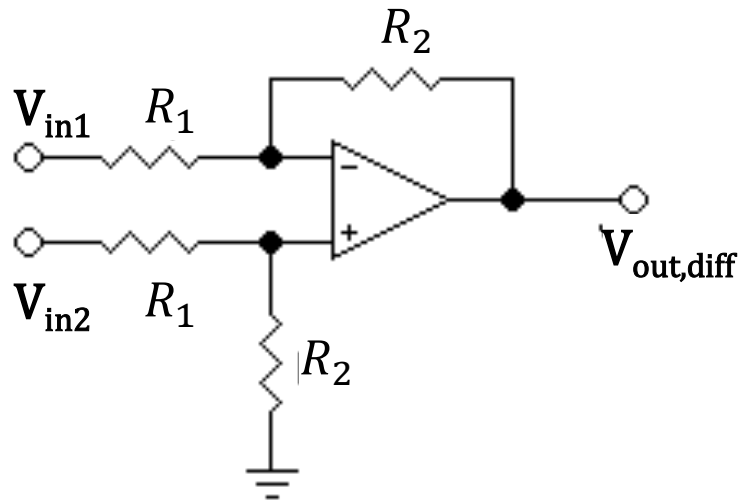
Input resistance into the inverting input terminal



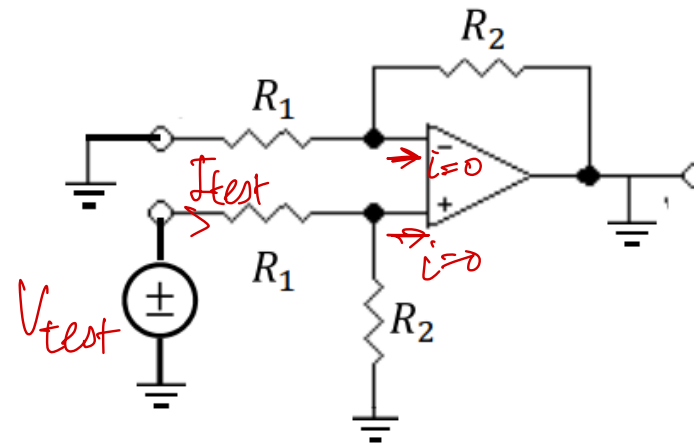
Input resistance for inverting input

$$R_{in} = \frac{V_{test}}{I_{test}} = R_1 + R_2$$

Issue of input impedance



Input resistance into the non-inverting input terminal

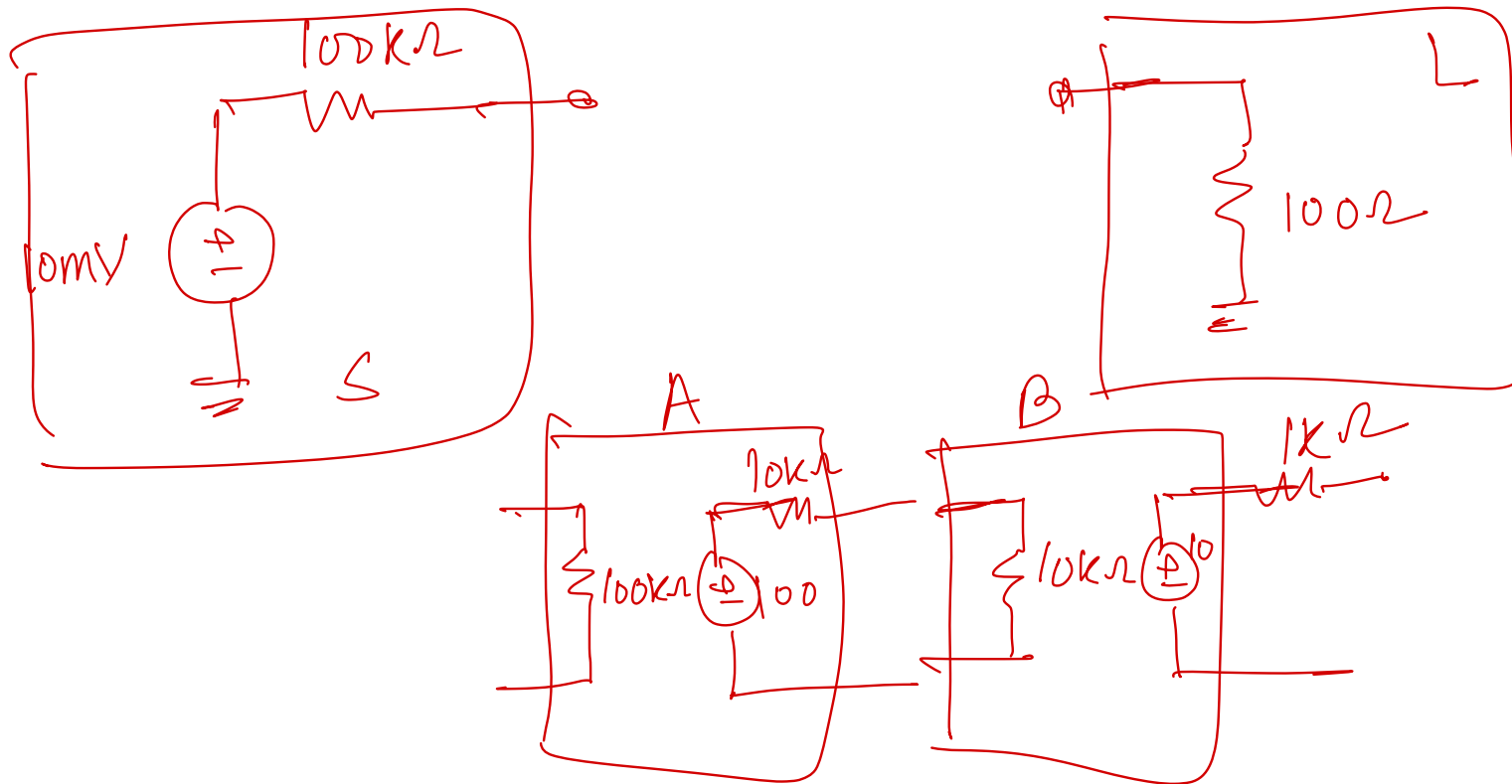


Input resistance at non-inverting input, $\frac{V_{test}}{I_{test}} = R_1 + R_2$

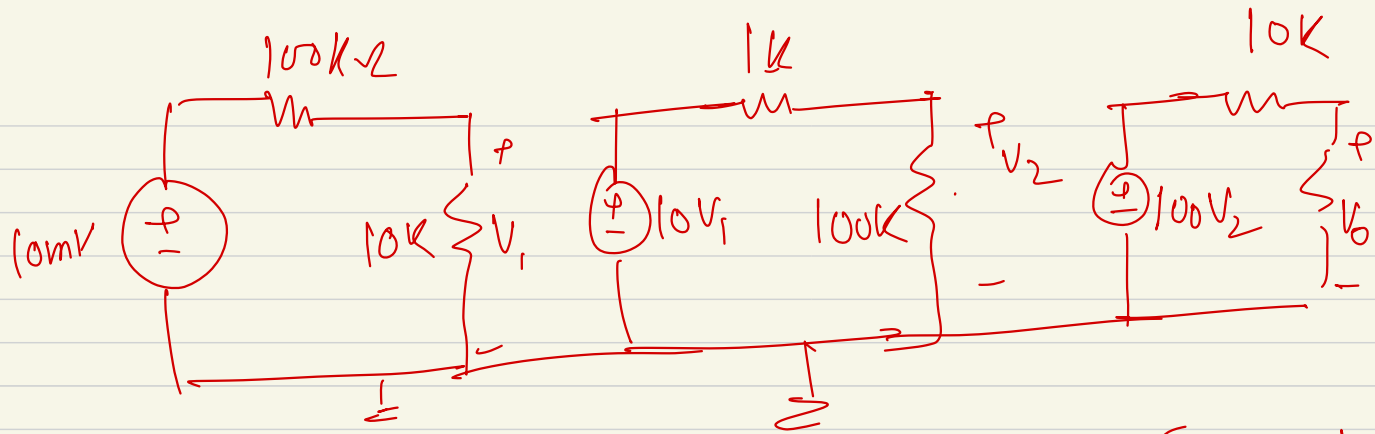
Practice Question to understand loading

You are given two amplifiers, A and B, to connect in cascade between a 10 mV, 100 k Ω source, and a 100 Ω load. The amplifiers have voltage gain, input resistance, and output resistance as follows: for A, 100 V/V, 100 k Ω , 10 k Ω , respectively; for B, 10 V/V, 10 k Ω , 1 k Ω , respectively.

Your problem is to decide how the amplifiers should be connected. To proceed, evaluate the two possible connections between source S and load L, namely, SABL and SBAL. Find the voltage gain for each both as a ratio and in decibels. Which is the amplifier arrangement the best?



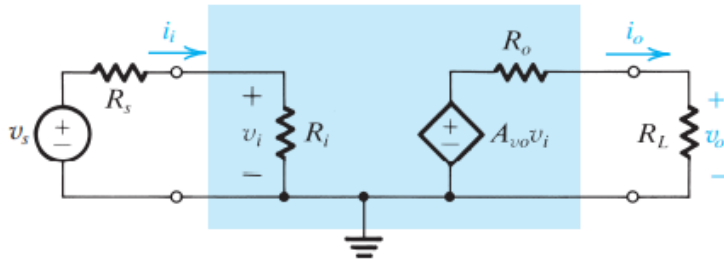
SABL
SBAL



SABL
$$U_0 = 10\text{mV} \times \frac{100\text{k}\Omega}{100\text{k}\Omega + 100\text{k}\Omega} \cdot 100 \times \frac{10\text{k}\Omega}{10\text{k}\Omega + 10\text{k}\Omega} \cdot 10 \cdot \frac{100}{100}$$

SBAL,
$$U_0 = 10\text{mV} \times \frac{10\text{k}\Omega}{110\text{k}\Omega} \cdot 10 \cdot \frac{100\text{k}\Omega}{101\text{k}\Omega} \cdot 100 \cdot \frac{0.1}{10.1}$$

Another example (Take Home)



Suppose, I don't particularly require a voltage gain as the signal from the source (v_s) is fairly strong. However, the source resistance (R_s) is higher than the load resistance (R_L).

A transducer characterized by a voltage of 1 V rms and a resistance of 1 M Ω is available to drive a 10- Ω load. If connected directly, what voltage and power levels result at the load? If a unity-gain (i.e., $A_{vo} = 1$) buffer amplifier with 1-M Ω input resistance and 10- Ω output resistance is interposed between source and load, what do the output voltage and power levels become? For the new arrangement, find the voltage gain from source to load, and the power gain (both expressed in decibels).

Ans. 10 μ V rms; 10^{-11} W; 0.25 V; 6.25 mW; -12 dB; 44 dB